



Figure 1: Scratch user interface

of the sprite is called a 'costume'.

User interface: Open Scratch (v1.4). The UI has three main sections—the block palette, the scripts area and the stage.

The block palette contains the collection of scripts in the form of blocks, which are classified into eight sections: *Movement, Looks, Sound, Pen, Control, Sensing, Operators, and Variables*.

The scripts area acts as the compiler in Scratch. The commands from *Block Palette* are dragged into *Scripts Area* and arranged for execution according to the logic. On clicking the sprite, the scripts are executed from top to bottom.

The stage is the final output screen. On executing (clicking the *Green Flag* button) the commands, compilation starts in the scripts area and the resultant output is displayed on the stage. The right-bottom corner of the stage displays the x-y coordinates of the mouse pointer with the centre of the stage as the origin. The list of all sprites is displayed in the *Sprites List*.

Constructs in C and Scratch

Let us compare a factorial program in C and Scratch.

Factorial program in C

```
#include<stdio.h>
int main()
{
    int num=0,fact=1;    //initialize num with 0 and fact
    with 1
    printf("Enter the number\n");
    scanf("%d",&num);
    if(num == 0)
        printf("%d",fact);    // if the number is zero, then
    factorial will be 1
    else
    {
        while(num>0)    // Loop to calculate factorial
        {
            fact=fact*num;
            num--;
        }
        printf("Factorial is %d",fact);
    }
}
```

```
return 0;
}
```

Factorial program in Scratch

The set block initialises the variables *fact* and *num* to 1 and 0 respectively. Ask block is used to display the statement 'Enter the number'. On pressing *Enter*, it automatically assigns the keyboard input value to the variable *answer*. The value in *answer* is copied to *num* using set block and calculations are carried out.

The coding is done with a few drags and drops of blocks. This unique feature makes Scratch a better platform for beginners in programming, helping them to enhance their skills as well as creativity.

Creating a game using Scratch

About the game

Sprites (objects): A car, gear, tank, obstacles and steering_wheel.

Variables: Points, a gear, gliding_time, fuel, time_gap and random.

The objective of the game is to ride the car as far as possible by avoiding obstacles (like cars, trucks, etc). The game starts when the user clicks the *Green Flag* button. As the car starts to move, the value associated with *Fuel* decreases. The gamer can control the motion as well as the speed of the car using steering wheel and gear sprites,



Figure 2: Stage and sprites